

The EngTrace evaluator pilot

September 2026. What we ran, what we learned, and how the full benchmark will be scored.

The pilot asked one question: *how should EngTrace evaluate reasoning at full scale, and is the published framework the right instrument?* Six candidate evaluators were run over a frozen slice of the benchmark, scored against labels from 15 domain experts, and stress tested with defects we planted ourselves. This is the result.

In short. The published framework spends its budget on LLM judges and reports a number that is mostly answer correctness, measured with a parser that is wrong on a quarter of traces. A deterministic stack scores the same traces more accurately for a fraction of the cost, and a judge is worth paying for in exactly one place: the steps no checker can verify.

1. How the pilot was built

1.1 The frozen slice

60 problems drawn from five engineering branches and three difficulty levels: 15 templates with four instances each. The slice is frozen with a recorded master seed and a manifest hash, and every scoring run re-derives it and compares bytes before it is allowed to score, so two runs months apart are known to have read the same problems.

Five models generated solutions, giving **300 traces**. Two smaller models were added as a robustness cohort, checked mechanically but not scored as results.

1.2 The six candidates

	what it computes	headline score
E0	the published framework: semantic step matching, then a panel of LLM judges on the steps it cannot match	share of reference steps recovered
E0-3J	E0 with its third judge connected, which in the published code was never actually called	same
E1	E0's method with all three judges replaced by families outside the evaluated suite	same
E2	three open process reward models scoring every step, run on local GPUs	fraction of steps above the reward threshold, and the minimum step reward
E3	deterministic milestone verification: did the trace state the quantities a correct derivation must reach, order free and unit aware	milestone coverage
E4	E3 plus a check of every arithmetic claim the trace displays	coverage, plus arithmetic consistency
E5	E3 first, then one judge only on the milestones E3 cannot settle	strict milestone coverage

All six scored the same 300 traces, resumable and keyed by a configuration hash, so a change in scoring cannot be confused with a change in data.

1.3 The expert labels

15 domain experts, three per branch. Every trace was labelled by three experts **of its own branch**, each producing:

- a label for every step: correct, alternative correct, incorrect, or not a claim, with an error type and a written reason when incorrect;
- a status for every milestone the template defines;
- a verdict on the final answer: correct, partial, incorrect, or not stated;

- an overall verdict on whether the reasoning was sound.

Traces were blinded behind opaque codes, assignment was fixed in advance, and a shared calibration set was labelled across branches so that agreement *between* branches could be distinguished from agreement *within* one. 1,020 submissions in total, about 13 hours of expert time.

1.4 Three quality rounds

These turned out to matter more than the labelling itself.

1. **A reason on every step called incorrect.** All 1,042 of them. Without this a disagreement cannot be adjudicated, only counted.
2. **Verification.** Each expert re-labelled seven of their own traces, blind to their first pass. This measures self-consistency, which is the ceiling that any between-expert agreement should be read against.
3. **Adjudication.** Every step where a branch's three experts split 2 to 1 went back to all three, blind, with the three original labels shown anonymised and shuffled so no reviewer could tell which was their own. The majority of that second pass became the label. 272 split steps across 134 traces.

Two of the fifteen label sets were re-annotated after verification identified them as least self-consistent. Both originals were kept.

1.5 How the comparisons were computed

- **Intervals.** 2,000 bootstrap resamples on every comparison, with each evaluator's difference from the published framework computed on the same resamples, so a paired comparison is not reported as two independent ones.
- **Clustering.** The 300 traces are 15 templates with four instances each, and instances of a template are the same problem with different numbers. Every headline comparison is therefore also reported with **templates** resampled rather than traces.
- **Power.** From the clustered standard error, the smallest difference detectable at 80% power and 95% confidence, which is what turns a null result into a statement about what the design can see.
- **Baselines.** Two: the framework's own answer check, and the experts' answer verdict, which is the best any answer check could do.

1.6 Planted defects

Two objections follow any result drawn from the labels. The arithmetic rule we tested is the same rule the annotation guide gave the experts, so their agreement is partly built in. And conceptual error behind a correct answer cannot be studied on this corpus at all: the experts found three such steps against 175 calculation slips.

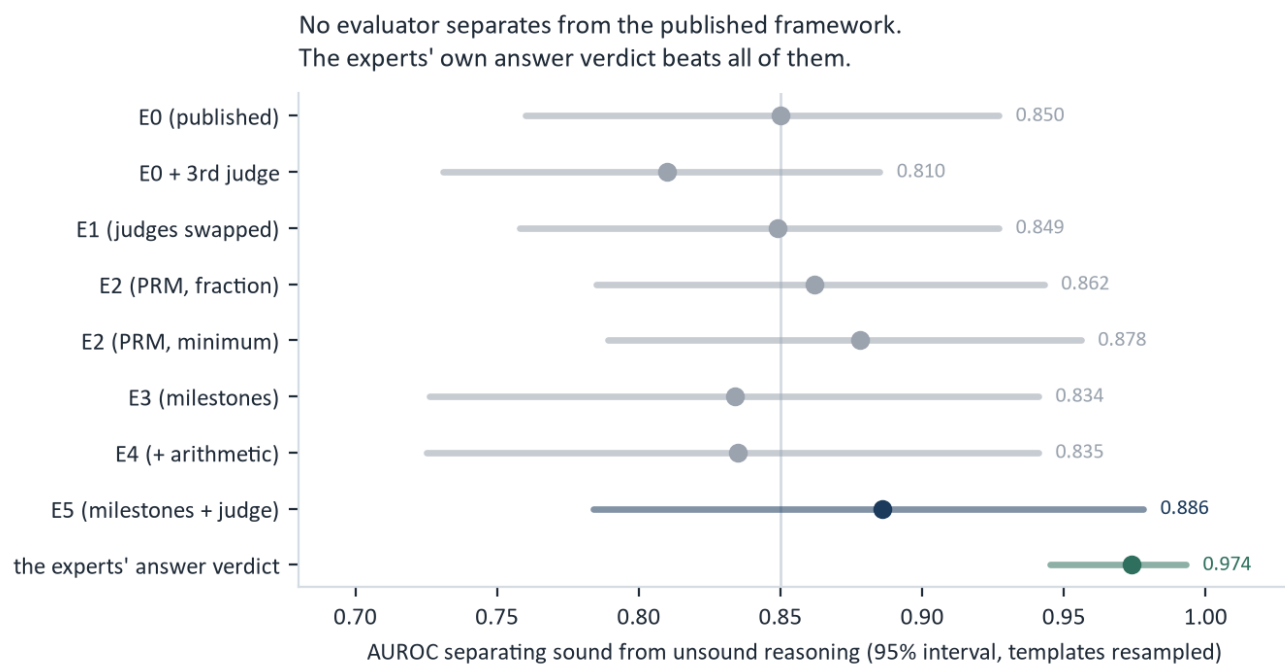
So we built a set whose ground truth is true **by construction**. From the 129 traces the experts called clean, each receives exactly one defect:

family	n	what changes	what is held fixed
arithmetic	60	one digit of one displayed intermediate value	later steps and the final answer
conceptual	60	the reasoning a step states: a criterion flipped, a rule misstated, a stated formula contradicting the one actually used	every digit in the trace
control	60	nothing	everything

Each plant is verified without the checker under test: an arithmetic plant is proved wrong by recomputing the claim at full precision or against the template's own milestone value, and a conceptual plant must leave the trace's digit string byte identical. The build is seeded and reproduces exactly.

2. What we found

2.1 The final answer decides the trace-level verdict



The experts' own final-answer verdict predicts their soundness verdict better than any evaluator does. Of 228 traces with a correct final answer, their holistic verdict calls only **three** unsound.

This is structural rather than incidental: **a trace-level score cannot rank reasoning evaluators**, because it mostly measures whether the answer is right. Everything that distinguishes the candidates happens at the step and milestone level, which is where the rest of the pilot works.

2.2 The published answer check is wrong on a quarter of traces

It disagrees with the experts on **72 of 300**, and almost always in one direction: 68 are traces the experts call correct and it calls wrong. In 45 of those, the correct value is present in the trace's own answer line.

Three defects cause it. The reference answer line often restates the question's inputs, and the first number on it is read as the answer. A third of the templates answer with a word rather than a number. And the reference value itself is taken as "the last number in the solution", which for one template is the exponent inside the unit m^3/s , so a trace was scored correct or wrong according to whether it wrote its units in ASCII.

answer accuracy	experts	published framework	corrected check
strongest model	0.950	0.617	0.950
all 300 traces	0.760	0.547	0.753

The benchmark understates every model by about 21 points and mis-ranks the top. The replacement is deterministic and needs no model:

1. read the trace's answer segment from its final-answer heading rather than its last "Answer" marker, so a multi-part answer is not truncated to its last part;
2. take each target from a quantity the reference **computed**, not from any number it prints, which removes restated inputs and formula constants without per-template rules;
3. score every part separately and return **correct, partial or incorrect**, because 19 of the 300 traces are genuinely partial and a binary check cannot represent them;

4. accept a value within a relative tolerance **or** within one unit of the last digit it displays, or of the reference's, since 114 for 113.55 is a correct rounding while 50 . 20 for 50.05 claims two decimals and gets them wrong;
5. allow unit factors, and read superscripts, LaTeX and fractions as the numbers they are.

It agrees with the experts on **0.947** of traces where the framework manages 0.747. Its one fitted parameter, the relative tolerance, was chosen on one half of the traces and reported on the other, both ways round.

2.3 Judge identity does not matter, so the panel is not buying accuracy

Replacing all three judges with families outside the evaluated suite moved the pooled score from 0.379 to 0.378. Connecting the never-called third judge moved it by +0.013, most of which proved to be a sampling artifact rather than the judge. This answers the independence objection and, at the same time, removes the argument for paying for a three-judge panel.

2.4 Milestones with a residual judge is the most accurate reasoning metric, and the cheapest

milestones, against the experts	precision	recall	F1
deterministic matching (E3)	0.926	0.915	0.921
deterministic, then a judge on the residue (E5)	0.930	0.989	0.958

Three quarters of milestones are settled with no model in the loop. Measured over the same 300 traces: **\$0.47 against the published framework's \$6.53**, a thirteenth of the cost, and more accurate.

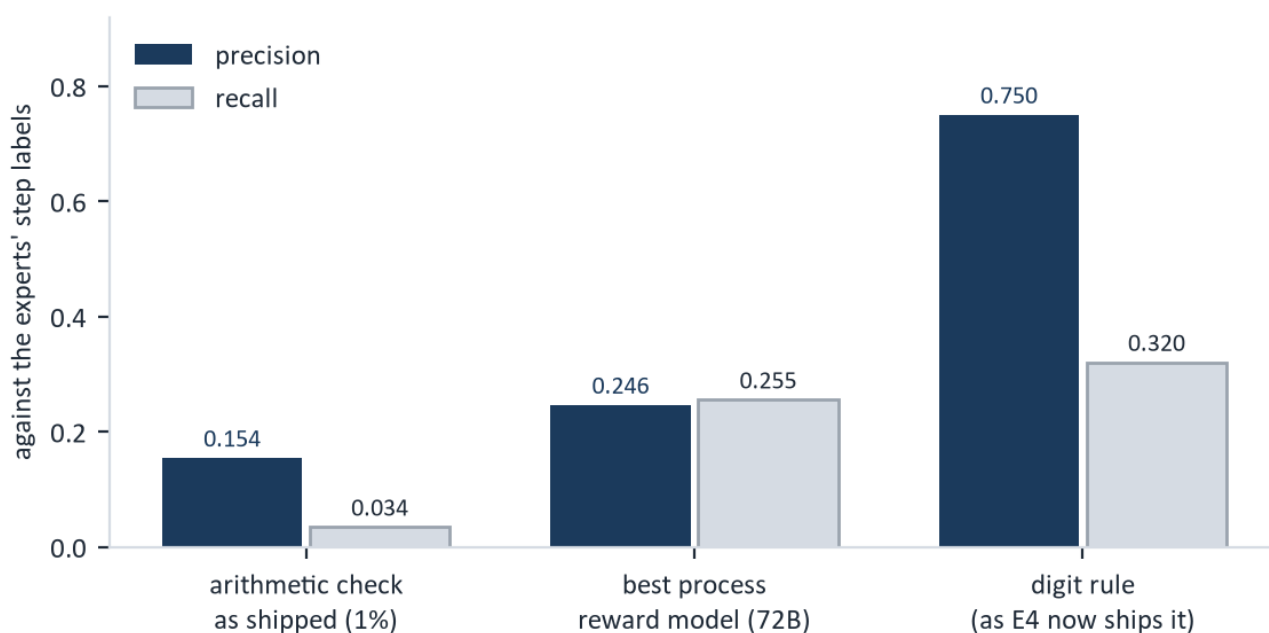
2.5 Process reward models over-flag, and the threshold is not the reason

The strongest of three open PRMs ranks steps well (AUROC 0.825) but only about half its flags are real errors, and inside correct-answer traces that falls to **0.246**. A multi-domain PRM found 7% of the errors and was dropped.

Calibrating the flag threshold on one half of the traces and reporting on the other gains **-0.006**: the default sits on a flat optimum. Inside correct-answer traces no threshold reaches precision 0.80 on held-out data at all, so the over-flagging is a property of the model rather than of the cut-off.

2.6 Arithmetic errors are deterministically detectable

Finding a wrong step inside a trace whose answer is right.
Three flags in four are real once the tolerance is the displayed precision.



The experts' own rule is mechanical: *rounding is not an error, a wrong digit is*. Applied by machine, it recomputes each claim from the numbers the trace itself shows and asks whether the displayed value is a correct rounding at the precision shown.

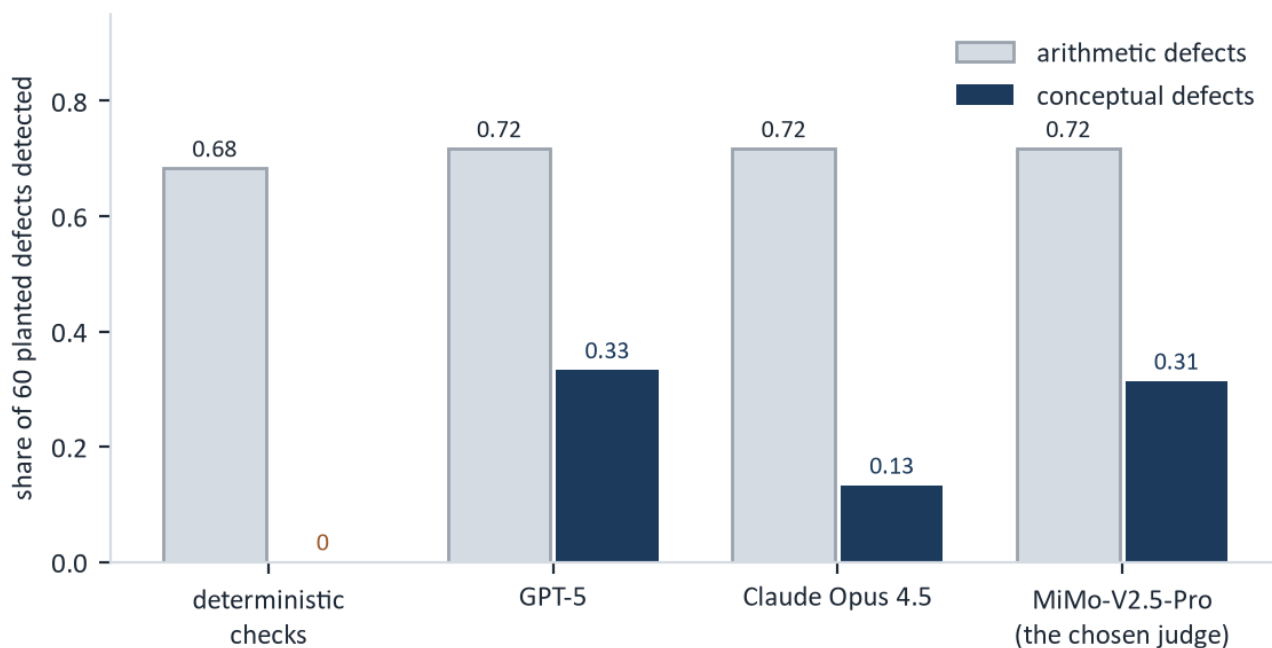
The framework already had this checker and was asking it the wrong question. Its 1% relative tolerance answers "is this number fabricated", which on this corpus is almost never, and it is blind to every slip the experts marked: at 1% it reaches **recall 0.034** on the steps that matter.

Two corrections were forced by re-validating on the reference solutions, whose arithmetic is correct by construction: compare a unit conversion after its unit factor, and do not hold a result to a precision its own displayed operands cannot pin down. Together they took the false-flag rate on reference solutions from 15.9% to **zero**, at a cost of five detections, all of them last-digit slips below one part in 100,000.

On planted arithmetic defects the rule scores **1.000 where it can parse the claim and 0.000 where it cannot**, so its recall ceiling is a parsing limit rather than a rule limit.

2.7 Nothing automatic catches a wrong idea behind a right answer, except a judge

Only a judge catches a misstated rule, and it catches about a third.
No false alarm on any untouched step, from any judge.



This is the pilot's central negative result, and the planted defects are what made it measurable rather than merely plausible.

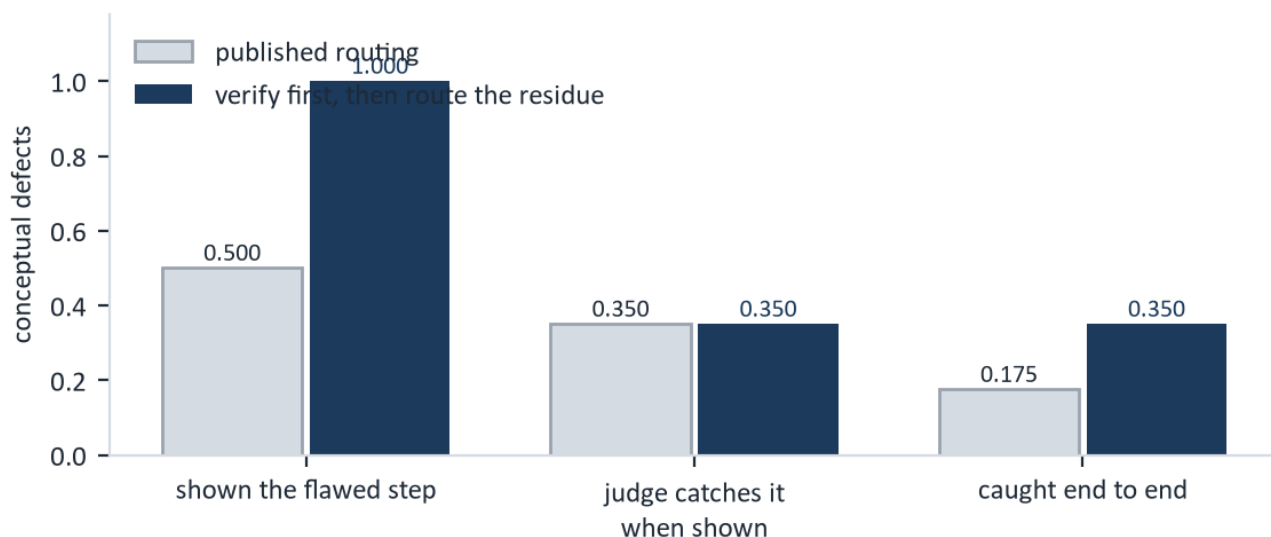
Three judges were probed on the framework's own tribunal prompt, with exactly one step under review. Each defect was judged **twice**: once in the planted trace and once in the same step of the untouched original. A judge that flags both is flagging the step, not detecting the defect, so only the difference counts.

	deterministic checks	GPT-5	Claude Opus 4.5	MiMo-V2.5-Pro
conceptual defects	0 of 60	0.333	0.133	0.314
arithmetic defects	0.683	0.717	0.717	0.717
false alarms on untouched steps	0.117	0.000	0.000	0.000

Two things follow. A trace that computes correctly, misstates the rule it is applying, and lands the right answer **passes every checker we have**. And the judge that satisfies the independence constraint performs like the frontier judges, so the full run does not have to choose between independence and sensitivity.

2.8 The framework rarely shows a judge the flawed step

The published framework shows a judge the flawed step as often as a clean one.
Routing what cannot be verified doubles what is caught.



Being able to catch a defect is not the same as being given the chance. Running the framework's routing for real on all 120 planted defects, with the judges stubbed so nothing was spent:

	conceptual	arithmetic
tribunal triggered	0.650	0.833
the flawed step is actually shown to a judge	0.500	0.767
<i>the same step, unmodified, is shown</i>	<i>0.500</i>	<i>0.683</i>
end to end (shown, then caught)	0.175	0.613

For a conceptual defect the routing carries **no signal at all**: the corrupted step reaches a judge exactly as often as the untouched one. It is forwarded because the matcher cannot match it, not because anything is wrong with it. The framework's judge spend buys a coin toss where a targeted router would buy a test.

A second finding fell out of the same run. The framework's answer check calls the final answer wrong on 32 of the 120 planted traces, although the plants never touch the answer. Those traces then enter its wrong-answer sample, so part of its judge budget today is aimed by a parser error.

2.9 The labels are reliable, and robust to replacing an expert

	value
between experts, step labels (Fleiss kappa)	0.781
within an expert, blind re-label (Cohen kappa)	0.828
between experts, milestone status	0.880
between experts, final-answer verdict	0.966

Self-consistency slightly exceeds between-expert agreement, which is the relationship you want to see: the labels sit close to the ceiling the experts' own consistency sets. Adjudication settled every split and changed 85 step labels, raising the count of steps called incorrect from 322 to 388.

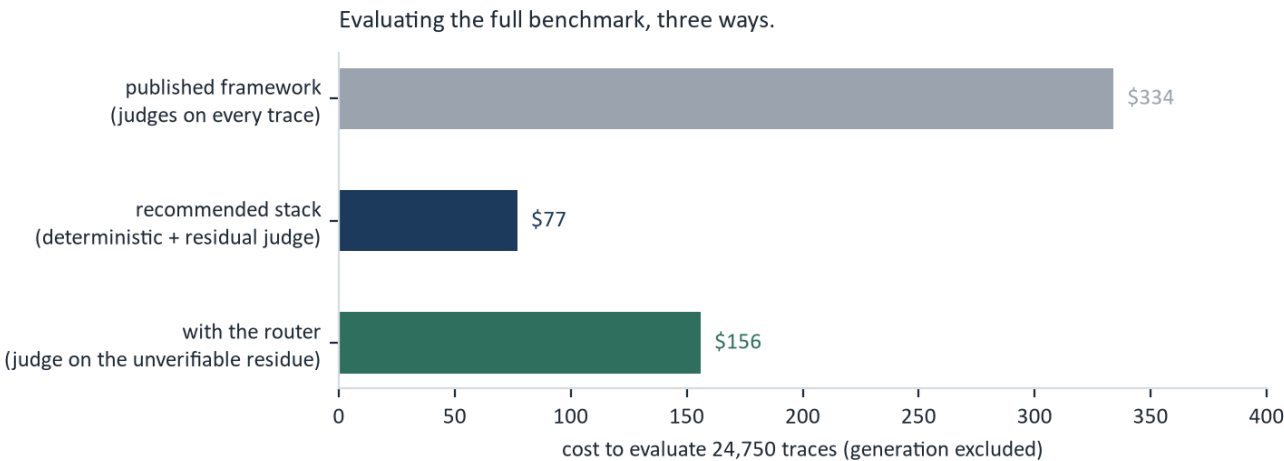
One stress test is worth repeating. **An entire expert's set, a fifteenth of the annotation effort and the least self-consistent one, was replaced, and not one ground-truth label moved.** Re-adjudicating the disputes that the replacement created then changed 12 of 2,091. The adjudication design is what absorbs a weak rater, and that can be shown rather than claimed.

2.10 300 traces are really 15 templates

Resampling templates instead of traces gives a design effect of about **3.7**: the slice is worth roughly **80 independent traces, not 300**. It can detect AUROC differences of about **0.12 to 0.19** between evaluators and no smaller.

Every null result in this pilot therefore means "no difference this large", not "no difference". The finding that clears the bar comfortably is the one that matters most: the dominance of the final answer, at +0.124 over the best evaluator with a clustered interval that excludes zero.

3. How the full benchmark will be scored



3.1 The stack

layer	how it works	model	cost
answer correctness	three-way check, targets taken from computed quantities	none	\$0
milestone coverage	deterministic, order free, unit aware	none	\$0
arithmetic integrity	the digit rule, at displayed precision	none	\$0
residual milestone judging	one judge, only on the milestones the deterministic pass cannot settle	MiMo-V2.5-Pro	~\$77
step routing	verify what can be verified, send only the residue to a judge	MiMo-V2.5-Pro	~\$79

About \$156 to evaluate the full benchmark, against roughly \$334 for the published framework, which we are not running at scale.

The router is in the stack because the pilot measured both sides of it. 63% of steps carry nothing a checker can recompute, 94% of traces hold at least one such step, and 14% of that residue is a step the experts call incorrect. Sending it to a judge lifts end-to-end conceptual detection from the framework's accidental 0.175 to about 0.35. It is the only component that addresses the pilot's central weakness, and it costs less than a quarter of what the published

framework spends to catch half as much. It does need building: the measurement exists, the implementation does not.

3.2 The models

The benchmark. Five branches x 30 templates x 15 instances = **2,250 problems** (870 easy, 870 intermediate, 510 advanced), answered once by each model on the roster.

The roster, 11 models, about \$403 to generate:

open weight (\$212.81)	frontier (\$190.18)
gpt-oss-20b, gemma-4-26b-a4b-it, deepseek-v4.1-flash, qwen3-235b-a22b, glm-5.3-flash, glm-5.3, muse-glimmer-30b, kimi-k3	gpt-5.4-mini, gemini-3.1-flash-lite, claude-sonnet-5

Every model the pilot generated traces with, or used as a judge, is excluded from the roster, so nothing being evaluated has also been an instrument of the evaluation.

The judge is MiMo-V2.5-Pro, for two reasons the pilot separated. It is independent: not a family on the roster, which is what the judge and judged objection requires. And that independence is free: on planted defects it matches the frontier judges (0.314 against GPT-5's 0.333 on conceptual defects, an identical 0.717 on arithmetic) with no false alarms, at about a third of a cent per call.

3.3 What is not being run, and why

	why not
the published tribunal as the reasoning metric	its score is dominated by answer correctness, and its parser is wrong on a quarter of traces
the third judge	moves the score by +0.013, most of it a sampling artifact
the judge swap as a main arm	already shown equivalent, 0.379 against 0.378; keep it as a robustness check on a sample
the multi-domain PRM	finds 7% of the errors
threshold tuning for the PRM	held-out gain of -0.006
the published framework at full scale	about \$334, and unlabelled full-run numbers would add cost without adding evidence, since the evaluator comparison belongs to the 300 labelled traces

The strongest process reward model can still be run on local GPUs at no API cost as a secondary per-step signal, about 15 GPU-hours for the full benchmark, but not as a headline number.

3.4 Two notes on reporting

Intervals. The full benchmark has 150 templates against the pilot's 15, so the same cluster-robust treatment applies with ten times the clusters, and the comparisons the design can support should be settled before the run rather than after.

The table will move. When the results are regenerated with the corrected answer check, **every model's accuracy rises by roughly 21 points and the ranking changes**. Readers will compare against the published version, so the paper needs a sentence explaining why.

4. Limits worth stating plainly

- **The slice is 15 templates**, five models, one of them weak. It separates frontier from weak; it cannot rank frontier models, and no evaluator claim should be read as one.
- **The hard-case finding is about arithmetic**. 175 of the 178 flawed steps behind a correct answer are calculation slips. Three are conceptual, which is why the planted set exists.

- **Planted defects are a diagnostic.** They say what an evaluator catches when a defect of a given shape is present, not how often models produce one. The conceptual defects vary in kind but not in phrasing, so they are not a held-out test.
 - **The digit rule shares its rule with the annotation guide,** so its agreement with the experts is partly by construction. The planted set is the independent check.
 - **The judges were asked about one step at a time.** A router would batch several into one prompt: more context per call, but also more steps competing for attention, and that is not measured.
 - **Expert labels are held outside the repository** by decision, so the analyses reproduce from the committed scripts only when the label files are supplied.
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5. Open items

1. **Template drift.** Three of the four deterministic layers derive milestones from the benchmark's templates, and a subset of the frozen items no longer reproduces byte identically. This must be fixed or formally pinned before the full run.
 2. **The router needs implementing.** Measured and costed, not built.
 3. **An anonymised ground-truth file** would let the analyses reproduce without exposing annotator data. Pending a compliance decision.
 4. **Two small follow-ups** from the label review: one probe step needs relabelling, and the annotation guide should say explicitly how to treat a notational slip whose computed result is right.
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Every number here is produced by a committed script, and every decision it refers to is recorded with its evidence in the project's decision log.